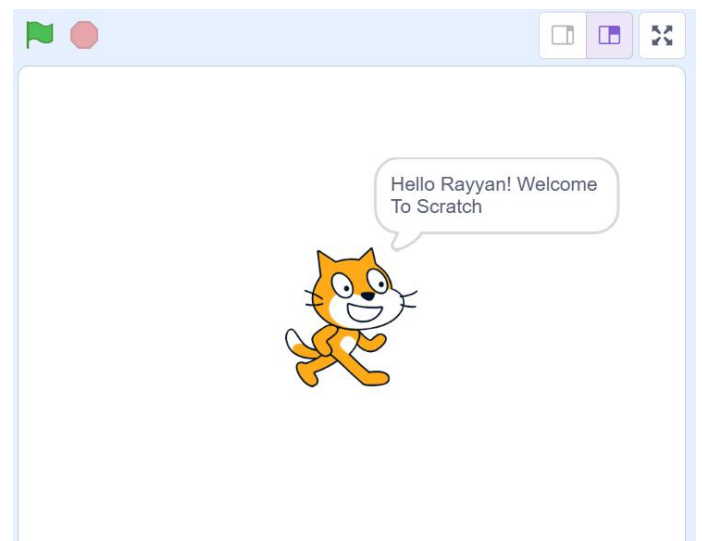
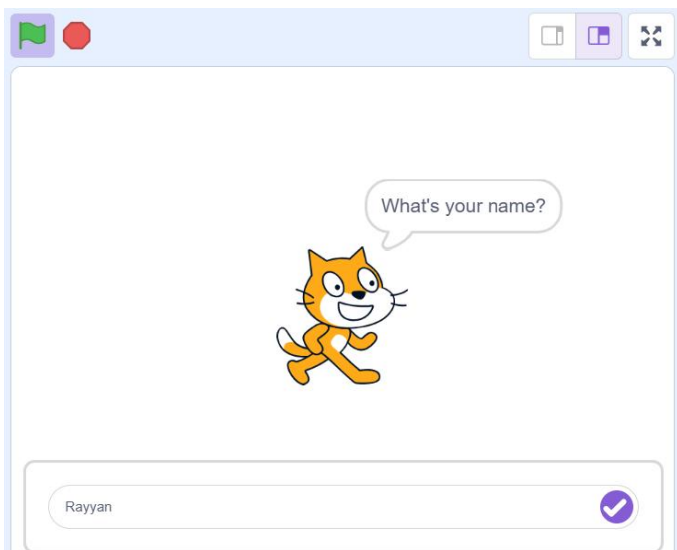
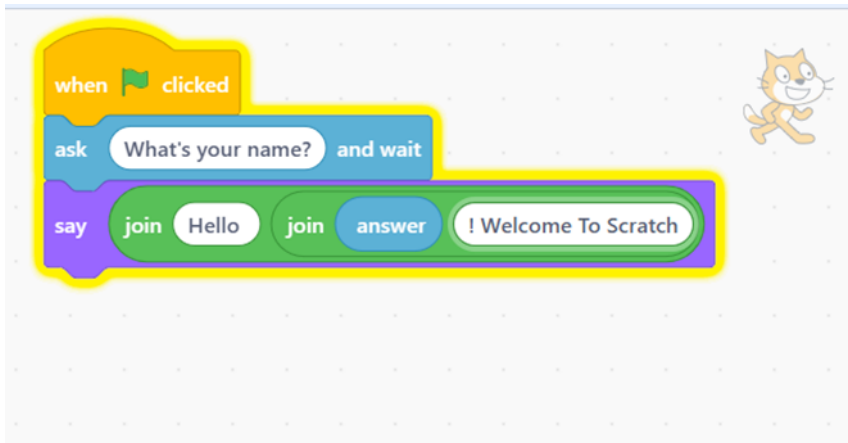


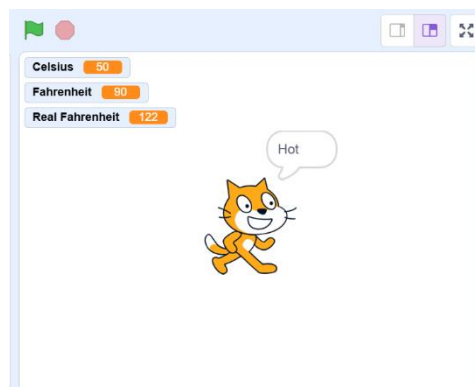
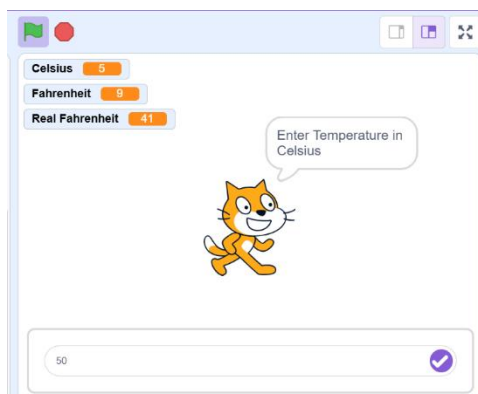
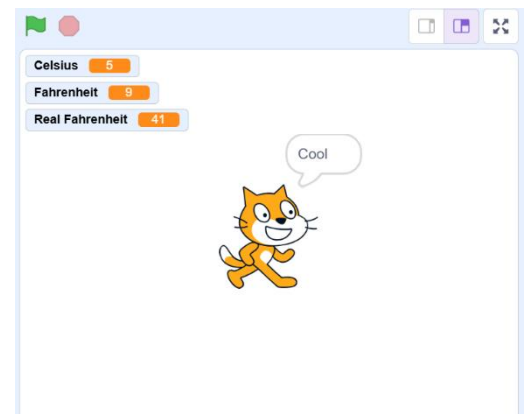
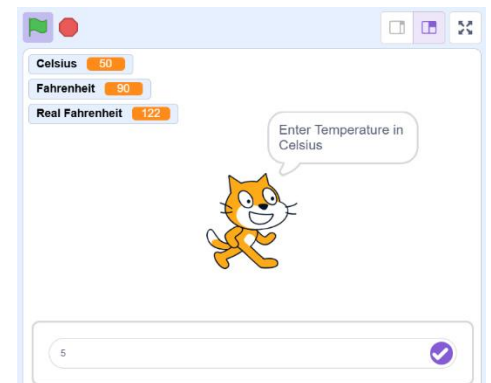
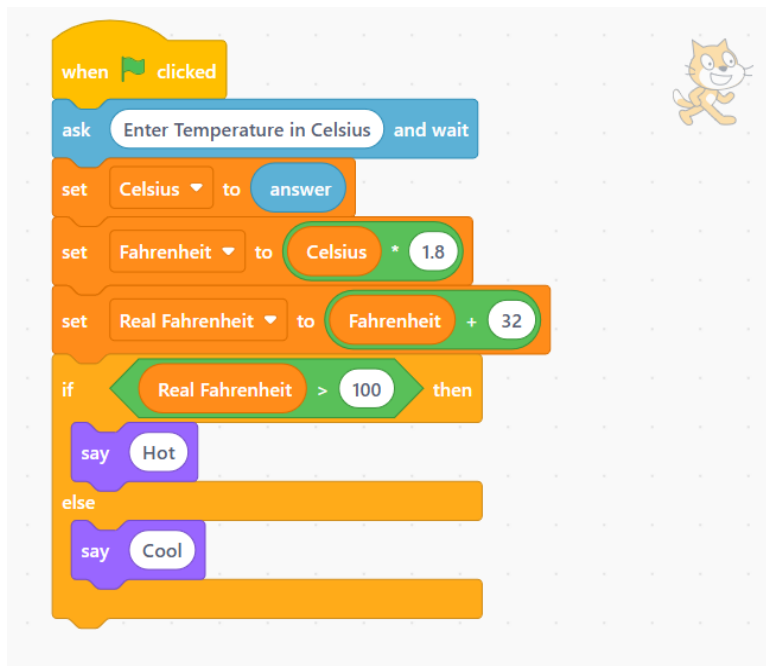
TASK # 01 — Personalized Greeting

Create a program where a sprite asks the user for their name and then gives a personalized greeting, such as Hello, [Name]! Welcome to scratch



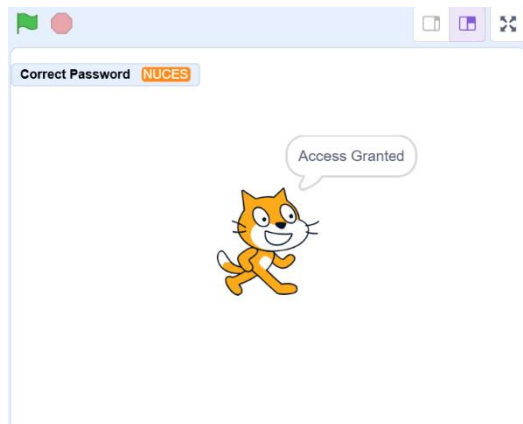
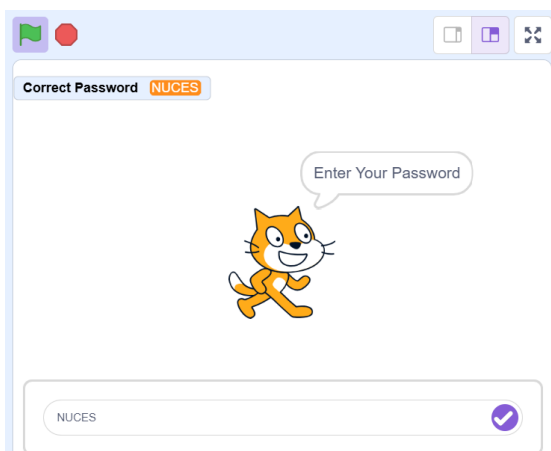
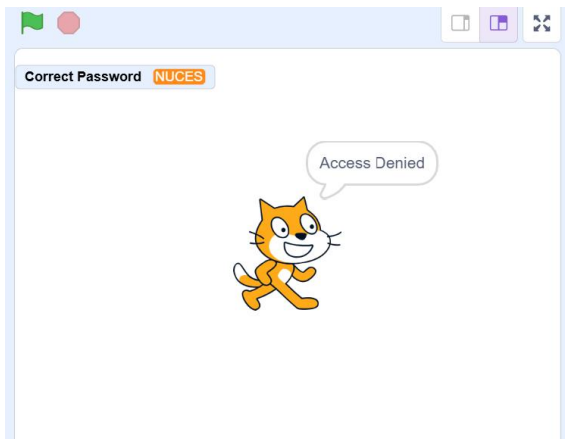
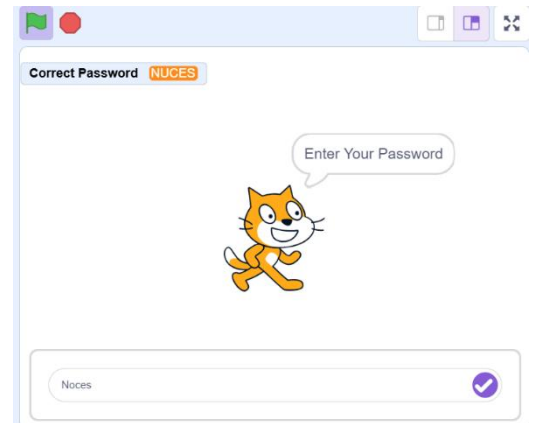
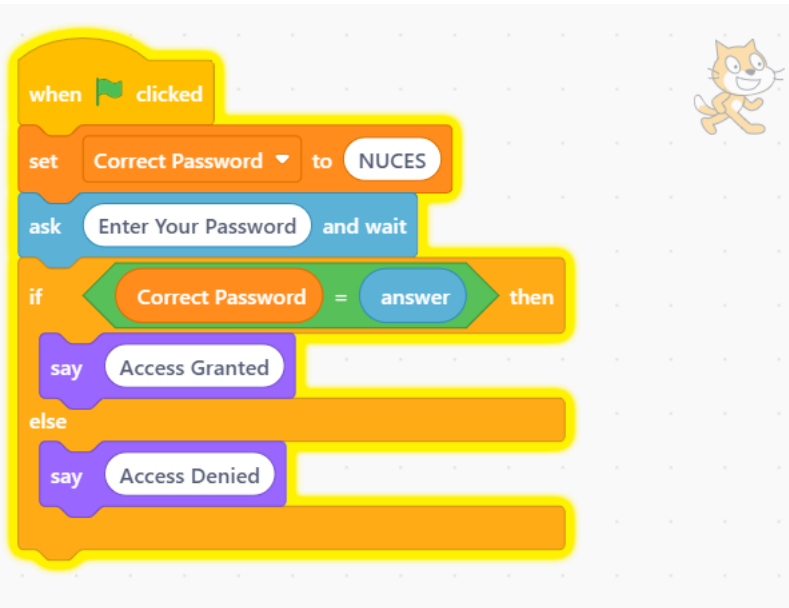
TASK # 02 — Temperature Converter with Feedback

Create a program that converts a user-input temperature from Celsius to Fahrenheit. Use a relational operator to check if the converted Fahrenheit temperature is above 100. If it is, the sprite should say hot otherwise, it should say cool.



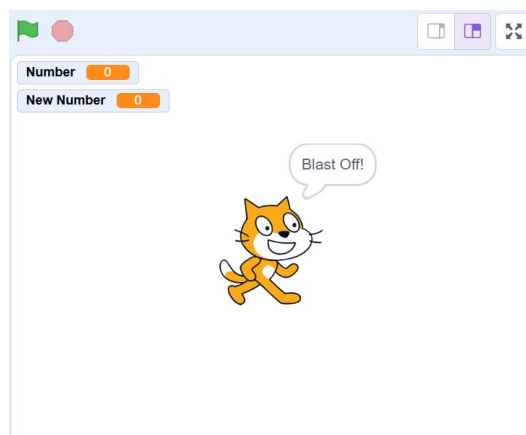
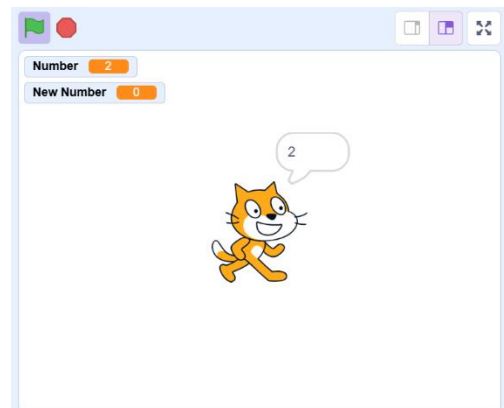
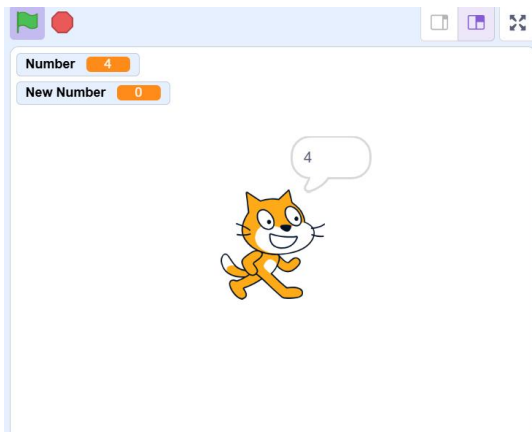
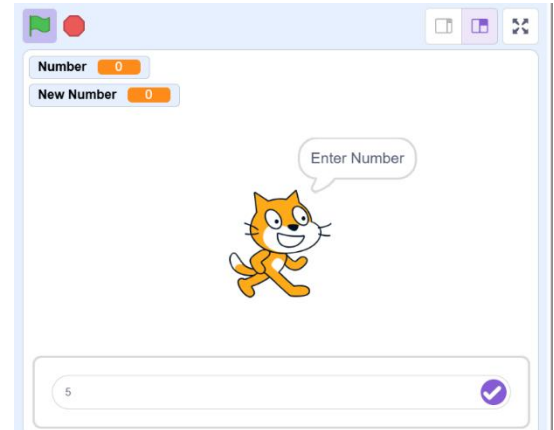
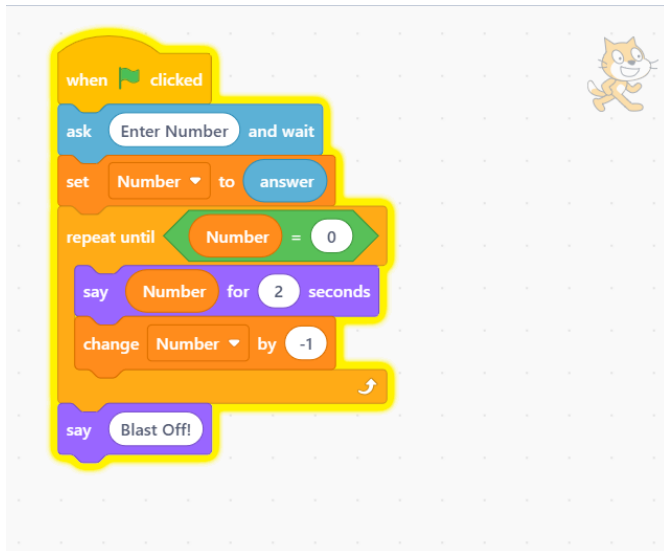
TASK # 03 — Password Checker

Create a program that stores a correct password in a variable (e.g. NUCES). Ask the user to enter a password. Use relational operators to check if the entered password matches the stored one. The sprite should say Access Granted if it matches and Access Denied if it does not.



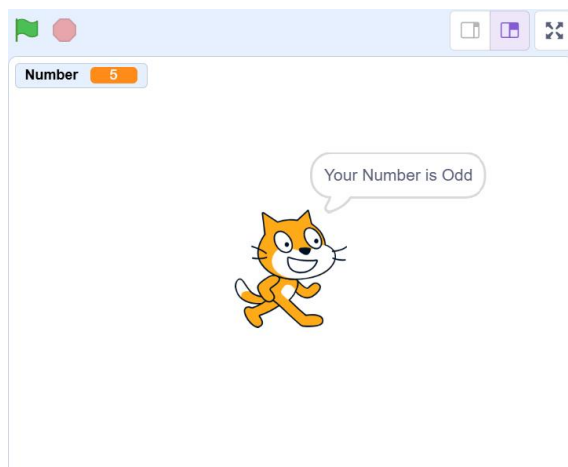
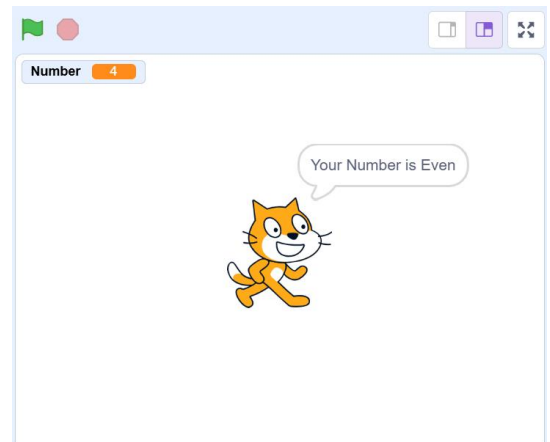
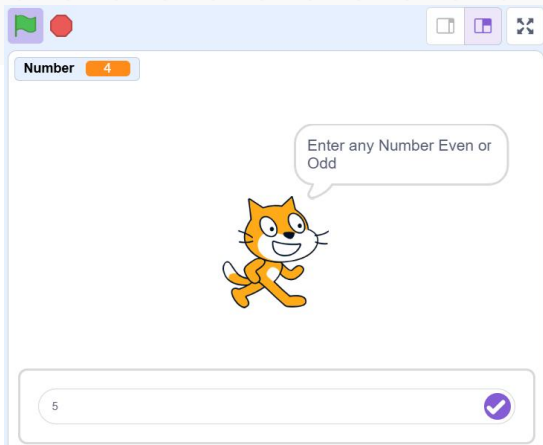
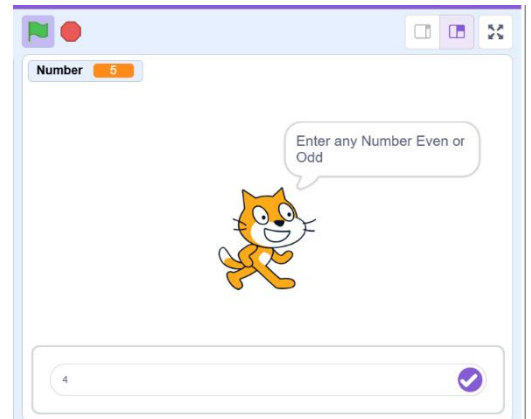
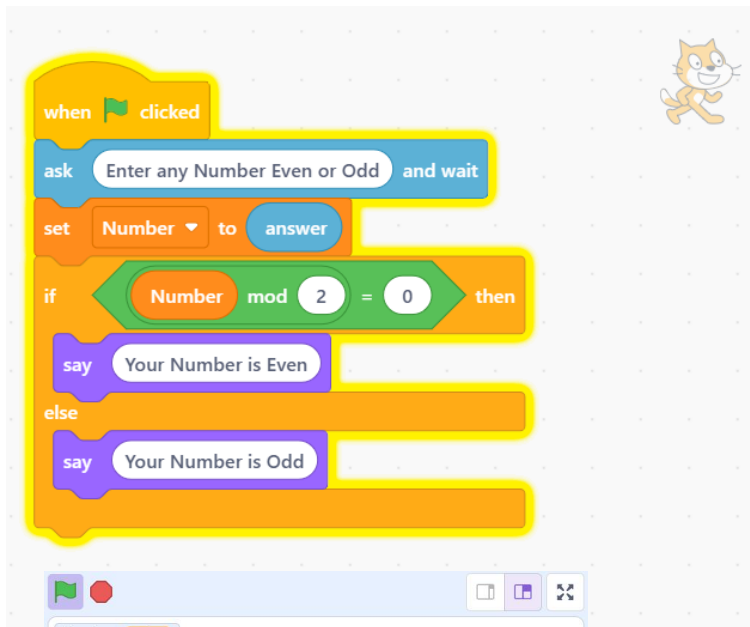
TASK # 04 — Countdown Timer

Create a program that asks the user for a starting number. Use a `repeat until` loop to make a sprite countdown from that number to 1, saying each number for one second. When it reaches 0, the sprite should say Blast Off.



TASK # 05 — Even/Odd Number Identifier

Create a program that asks the user for a number. Use an operator (hint: the `mod` block found in Operators) to determine if the number is even or odd. The sprite should then report the result (e.g., your number is even).



TASK # 06 — Sum of a User-Defined Range

Ask the user for a starting number and an ending number. Use a loop to calculate the sum of all whole numbers from the start to the end (inclusive). Display the final sum.

The image displays the Scratch code for calculating the sum of a user-defined range, along with three screenshots of the program's execution.

Scratch Code:

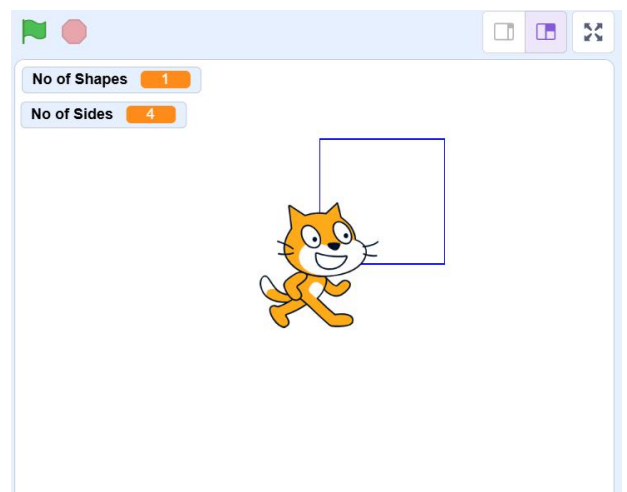
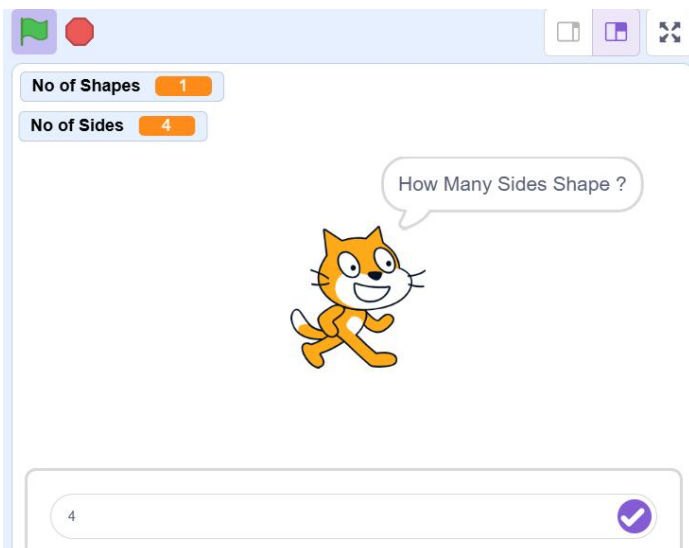
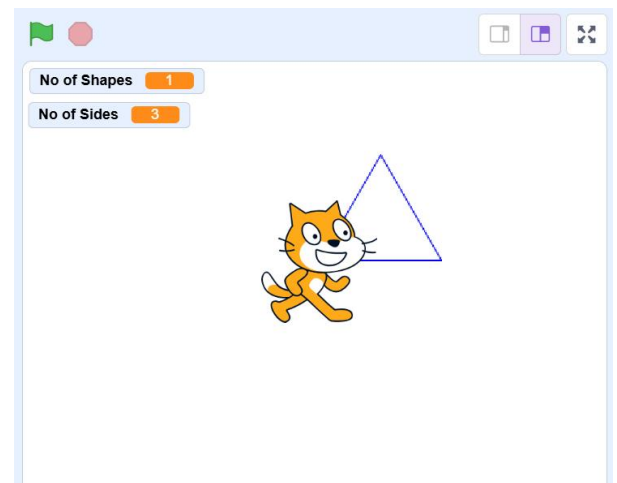
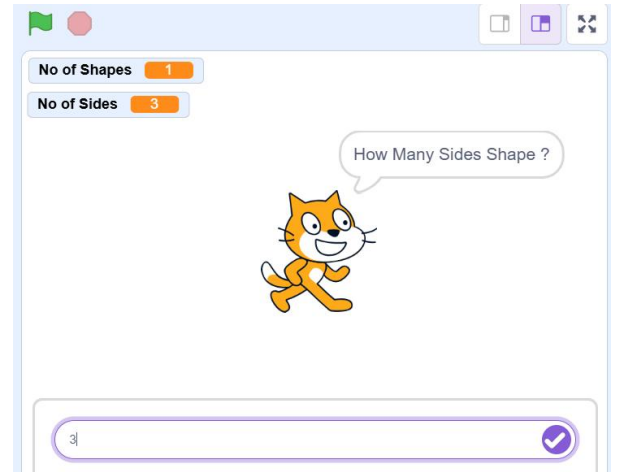
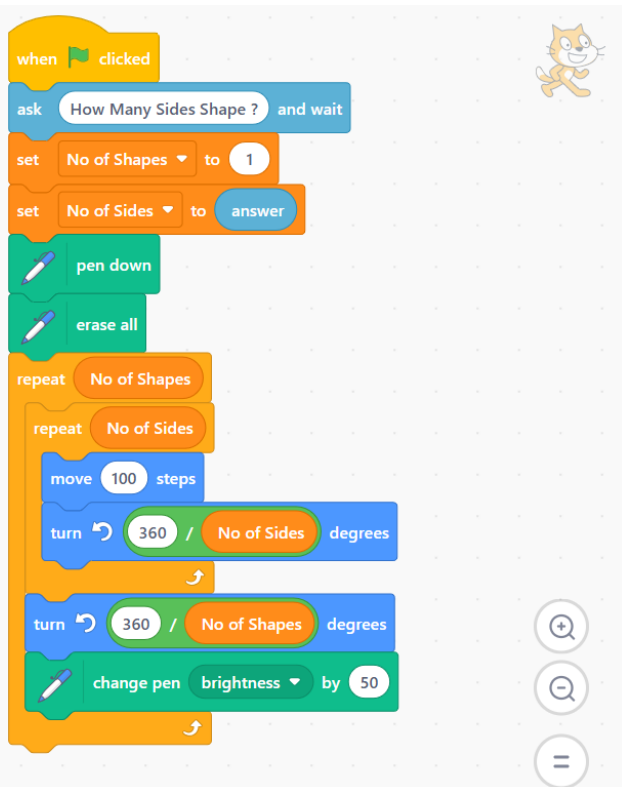
- when green flag clicked
- ask "Enter Starting Number" and wait
- set Starting Num to answer
- ask "Enter Ending Number" and wait
- set Ending Num to answer
- set Sum to 0
- repeat until Starting Num > Ending Num
 - change Sum by Starting Num
 - change Starting Num by 1
- say join "The Sum is" Sum

Execution Screenshots:

- Screenshot 1:** The program starts with Starting Num = 11, Ending Num = 10, and Sum = 55. A speech bubble asks "Enter Starting Number". The input field contains "11".
- Screenshot 2:** The program continues with Starting Num = 1, Ending Num = 10, and Sum = 55. A speech bubble asks "Enter Ending Number". The input field contains "10".
- Screenshot 3:** The program displays the final result: "The Sum is 55".

TASK # 07 — Interactive Shape Drawer

Create a program that asks the user how many sides they want a shape to have (e.g., 3 for triangle, 4 for square). Use a `repeat` loop with that value to draw the corresponding shape using the pen tools.



TASK # 08 — Number Guessing Game

Create a game where the program randomly chooses a number between 1 and 10 (use the `pick random` block from Operators) and stores it in a variable. Ask the user to guess the number. Use decision structures to tell the user if their guess is too high, too low, or correct. Use a loop to let the user keep guessing until they get it right.

